



DEALER BEST PRACTICE SERIES

Wireless Pager System
Optimizes Workflow
Process

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Wireless Andon Pager System Optimizes Workflow
Process 0

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1.0 Introduction

During the repair process in the CRC machining center, workflow interruptions and delays can occur due to a variety of reasons, such as issues with tooling, parts, logistics, technicians, etc. Quick and effective response by shop management to resolve the interruptions maintain the workflow and minimizes downtime of equipment and personnel, reducing the cost of downtime and improving the repair Turn Around Time (TAT).

2.0 Best Practice Description

The Ferreyros CRC machining center uses a wireless system to alert quality inspectors and supervisors of any problem in the production workflow process. This gives the operator the ability to call for the required assistance at the moment a problem occurs. For this purpose, the technicians need a system that allows instant communication with the proper resource to solve the problem and allow the continuity of production.

The machining workstations also require the transport of large parts into and out of the workstation. The wireless system is used to alert the parts runners when a job is completed to removed the completed component and bring in the next job.

This Best Practice presents a wireless system of alerts that allows direct communication between the workstation operator and the supervisor, quality inspector, or runner who will attend the call.

The system is set up so that each workstation has a wireless paging device that allows the operator to page the Supervisor, Quality Inspector, or Runner.

The team receives the call through a wireless wrist band receiver indicating attention is needed and which work station.

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**Old system**

Andon call lights

Work Station & Team Work Communication System

New system

Call button and wrist pagers

Work Station

RUNNER's Call Button

**RUNNER**

RUNNER's Close Button



INSPECTOR's Call Button

**INSPECTOR**

INSPECTOR's Close Button



SUPERVISOR's Call Button

**SUPERVISOR**

SUPERVISOR's Close Button



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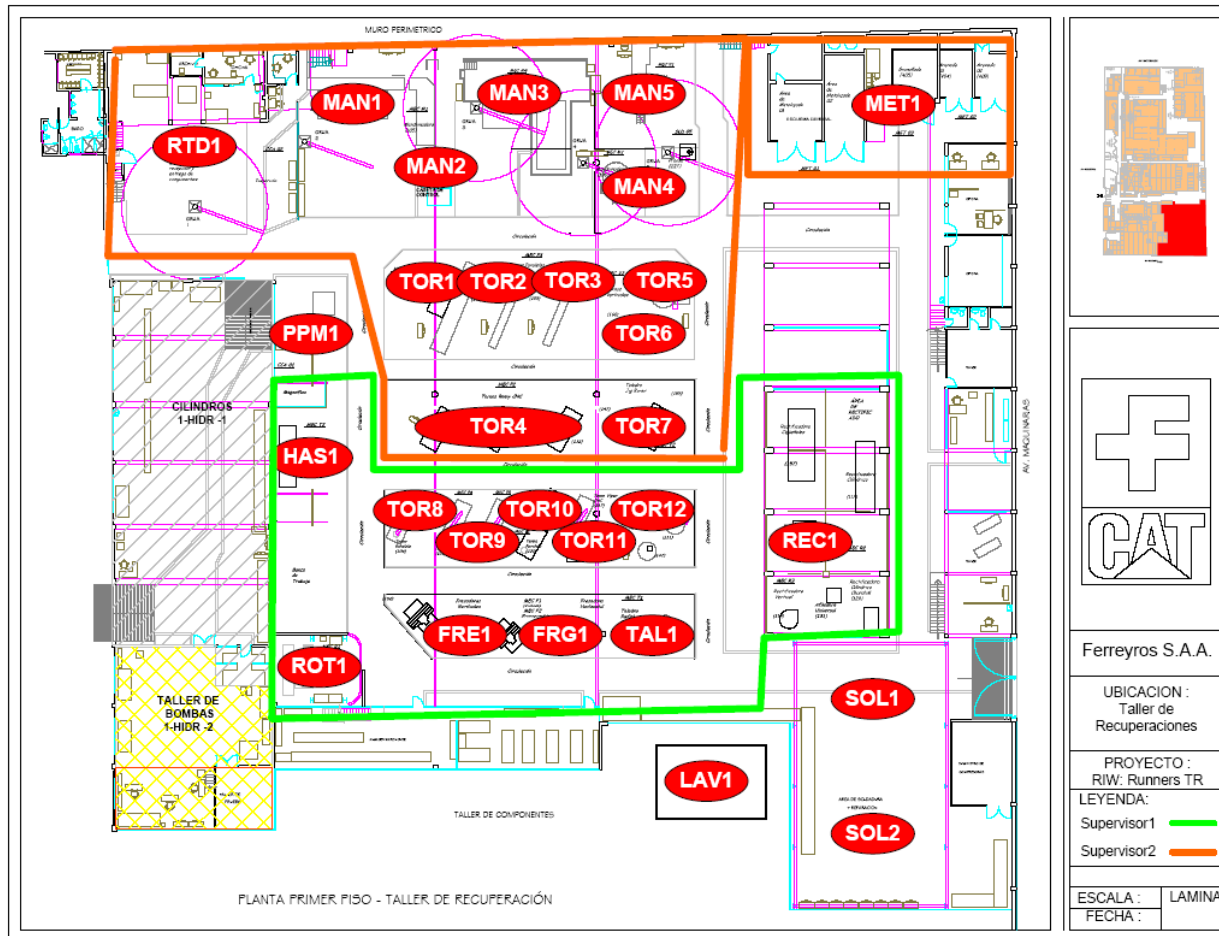
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To achieve efficient attention to the workstations, the workshop has been zoned according to load balancing. Pagers in each area are set up to notify the supervisor, quality inspector, or parts runner for that area.



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The system used by Ferreyros has software that stores all call information and attention to workstations by different teams, thus allowing different reports on the call time for staff, per workstation, per shift, etc.

The supplier for Ferreyros was Vellux, however, there are many commercially available suppliers of paging systems.

E-mail: velluxperu@promhispanos.com

Web Site: www.velluxsystems.com

The screenshot shows a web application titled 'Estadísticas de Vellux'. It features a sidebar menu with options like 'Reporte de Puestos', 'Reporte por Actividad', and 'Reporte por Puesto'. The main area displays a report for 'Puesto 1' with filters for 'Turno' (Unico), 'Fecha' (05/17/2010 to 05/25/2010), and 'Hora'. A table lists activities with columns for 'Llamada al', 'Desde', 'Hasta', 'Tiempo de respuesta', and 'Zona'. The table shows several entries for 'SUPERVISOR' and 'RUNNER' across different zones like 'Zona 110', 'Zona 103', and 'Zona AB1'.

Llamada al	Desde	Hasta	Tiempo de respuesta	Zona
SUPERVISOR	2010-05-18 11:29:43	2010-05-18 11:30:19	00:36	Zona 110
RUNNER	2010-05-18 11:24:25	2010-05-18 11:31:03	06:38	Zona 103
SUPERVISOR	2010-05-18 11:30:08	2010-05-18 11:31:05	00:57	Zona AB1
SUPERVISOR	2010-05-18 11:42:26	2010-05-18 11:42:29	00:03	Zona 110
SUPERVISOR	2010-05-18 11:42:20	2010-05-18 11:42:35	00:15	Zona 103
RUNNER	2010-05-18 11:42:37	2010-05-18 11:42:40	00:03	Zona 132
SUPERVISOR	2010-05-18 11:41:45	2010-05-18 11:43:22	01:37	Zona AB1

3.0 Implementation Steps

- Define the work area zone and number or work stations in the zone
- Define the responders for each zone: Supervisors, Quality Inspectors, Runners
- Acquire hardware and software and install
- Train personnel
- Implement

The wireless system project was conducted as Caterpillar Production Systems project to reduce excess motion, time, over process, inventory, and transport.

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4.0 Benefit

- Communicate a delay in seconds to Supervisors, Quality Inspectors, and Runners
- Lets you route the call to the people assigned to each point
- Reduces travel time and increases shop productivity, which reduces cost and increases profitability
- Vibrating wristbands alert responder no matter where he may be located. Much more efficient than the “visual” Andon light system
- Reduce the risk of accidents. The workers are not distracted by looking at traffic lights
- The workshop can measure delay times and number of calls of each operator, optimizing time and human resources work. This will allow optimal management within each workshop
- More organized workshop and good communication

5.0 Resources Required

- Call buttons for each workstation for every workstation
- Pager wristband for each Supervisor, Quality Inspector or Runner
- Transmission master VM2700 multi functions
- 2 relays (01 repeater VBR-100H and 01 repeater VPR-100H)
- Custom software
- Total Investment US\$ 8,200 for 35 workstations and 8 pagers.

6.0 Supporting Attachments / References

7.0 Related Best Practices

8.0 Acknowledgements

This Best Practice was written by:

Name Patricia Valdivieso Naupari
 Title Black Belt CPS & 6 Sigma
 E mail pvaldivie@ferreyros.com.pe
 Phone 00 51 1 6265101

Recognition:

Team: Liliana Falcón/Planning Chief
 Rafael Volta/Machine Shop Chief
 Gabriel Sedano , Nelson Chappa /Planners
 Supervisors, Quality Inspectors and Runners
 Sponsor: Carlos Dongo/ Manager CRC
 Owner: Rodolfo Gerstein/ Manager Machine Shop
 Black Belt : Patricia Eguchi , Patricia Valdivieso

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